CSE STUDY HUB

DATA COMMUNICATION & NETWORKING UNIT 6

Research-Oriented Network Applications

SYLLABUS-OUTLINE GUIDE · NOT VERIFIED AGAINST CLASS NOTES

No Unit 6 class-note PDF was included in the reference files uploaded for this update. This guide follows the existing study-hub syllabus outline only. Confirm exact details with your teacher's handout before exam preparation.

Topics in the current outline

- Next-generation networks
- Software Defined Networking (SDN)
- Network Function Virtualization (NFV)
- AI-driven networking and cyber-physical systems

Exam answer structure

Define the technology → draw its architecture → explain how it works → give one application → discuss benefits, limitations and evaluation metrics.

UNIT 6 · Next-generation networks

High-speed networks aim to increase capacity, reduce delay and support more connected devices and services.

When studying a next-generation network, identify its architecture, performance goals, deployment constraints and security considerations.

This is a syllabus-outline guide only; the uploaded class reference set contains Units 1–5, not Unit 6.

UNIT 6 · Software Defined Networking (SDN)

SDN separates the control plane, which decides how traffic should be handled, from the data plane, which forwards packets.

A logically centralized controller can program network devices through software interfaces. This can simplify policy changes and network management.

For an exam answer, draw controller → switches/routers and label control-plane and data-plane roles.

UNIT 6 · Network Function Virtualization (NFV)

NFV implements network functions as software rather than requiring each function to run on dedicated proprietary hardware.

Examples of virtual network functions include firewalling, routing, load balancing and intrusion detection.

Compare NFV with traditional appliance-based deployment in terms of flexibility, scaling, cost and operational complexity.

UNIT 6 · AI-driven networking and cyber-physical systems

AI-driven networking can use data to detect anomalies, forecast traffic demand and support network optimisation. Explain the input data, model decision and resulting network action.

Cyber-physical systems combine computation, communication and physical processes. Network delays, reliability and security can affect real-world control.

Use a concrete case study and state the metric used to evaluate the system, such as latency, packet loss, throughput or detection accuracy.

QUICK REVISION & PRACTICE

SDN separates network control from packet forwarding. NFV implements network functions as software. AI-driven networking applies data-driven techniques to network operation. Cyber-physical systems connect digital computation with physical processes.

Self-test

- Explain SDN control and data planes.
- What is NFV? Give examples.
- How can AI support network management?
- Why does a cyber-physical system need reliable networking?
- Suggest metrics for evaluating network optimisation.